Eunsu Yun

Computer Vision · On-Device AI · Physical AI

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EDUCATION

Mar 2024 – Feb 2026	**M.S. in Computer Science**, Sangmyung University — Computer Vision Lab (CCLAB) · GPA 4.5/4.5 Thesis: A Study on a Lightweight Deep Learning Model for Classifying Normal and Abnormal Vasculature in Organoid Images
Mar 2020 – Feb 2024	**B.S.**, Sangmyung University — Convergence Electronic Engineering & Human Intelligence Information Engineering (Double Major) · GPA 4.16/4.5

PUBLICATIONS

2026 · Journal	C. Xie, J. E. An, H. J. Bae, **E. S. Yun**, K.-C. Ko, D. S. Jeong, O. S. Kwon, "On-Device Artificial Intelligence for Image Analysis in Resource-Limited Settings: A Review," *Applied Science and Convergence Technology*, 35(1), 1–14, 2026. co-author (4/7) · doi:10.5757/ASCT.2026.35.1.1
2026 · Journal	**E. Yun**, J. Kim, D. Jeong, "Lightweight Deep Learning Model for Classification of Normal and Abnormal Vasculature in Organoid Images," *Sensors*, 26(1), 112, 2026. SCIE · first author · doi:10.3390/s26010112
2025 · Conference	"Implementation and Processing-Time Analysis of an NPU-based Real-Time Iris Recognition System," IEIE Summer Annual Conference 2025. co-first author (1/7) · in Korean
2025 · Conference	"Hybrid Iris Recognition for Authentication Systems in AI Edge Device (NPU) Environments," ICGHIT 2025. co-author (2/5)
2024 · Conference	"Development of an AI-based Advertising Cost Prediction Algorithm," KIPS ASK 2024. co-author (4/8) · in Korean
2023 · Conference	"A Study on Deep Learning-based Banknote Serial Number Recognition," CEIC 2023. first author · poster · in Korean
2023 · Conference	"Proposal of an Ontology Generator Based on a Neuro-Symbolic Architecture," KIPS ACK 2023. co-first author (1/6) · in Korean

RESEARCH PROJECTS

Jun 2025 – Jan 2026	**Lightweight Organoid Vasculature Classifier** — M.S. thesis • Compressed EfficientNet (SiLU→ReLU, SE blocks removed); FP16 on Jetson Nano, INT8 on FuriosaAI Warboy NPU • ~40% smaller model, 99% accuracy on NPU, 31% lower inference latency; published in *Sensors* (SCIE)
Sep 2024 – Jan 2025	**In-Vehicle Face Recognition & Emotion Analysis** — industry project (DeepInsight) • Integrated YOLOv5 face detection, HRNetV2 landmarks, ArcFace+PCA identification, EmoNet/SVM emotion on one edge device • Face detection mAP50 0.99, identification accuracy 99%, ~40% smaller total model size
May 2024 – Aug 2024	**NPU-based Real-Time Iris Recognition System** — industry project (Hyundai MIB International) • YOLOv8s segmentation + iris-code (Hamming distance) hybrid; deployed on Jetson Nano and Mythic NPU • Mean IoU 0.96; 47 ms (21 FPS) end-to-end on NPU; collaborated in English with an overseas technical team
Oct 2023 – Dec 2023	**Banknote Serial Number Recognition** — undergraduate research • Built a 12,998-image dataset (incl. Cyrillic); ResNet + BiLSTM CRNN with CTC loss; 99.15% accuracy
Mar 2022 – Jun 2022	**Camera-based ADAS** — course project • OpenCV lane detection + YOLOv3-tiny vehicle detection on Jetson Nano (~23 FPS); LDW and forward warning

EXPERIENCE

Mar 2024 – Feb 2026	**Graduate Researcher**, Computer Vision Lab (CCLAB), Sangmyung University
Jun 2023 – Feb 2024	**Undergraduate Researcher**, Computer Vision Lab (CCLAB), Sangmyung University
Apr 2024 – Dec 2024	**Arduino Coding Club Instructor**, Sangmyung University Girls' High School

AWARDS · CERTIFICATIONS · TRAINING

Sep 2023	KIGAM President's Award — Encouragement Award, Geoscience Data & AI Competition (Hackathon track) — LSTM anomaly detection, F1 ≈ 96%
2026	Engineer Information Processing (Sep) · SQL Developer, SQLD (Jun) · ADsP (Jun) · TOEIC Speaking IH (150, Jun)
Oct 2026 – Apr 2027	NVIDIA AI CAMPUS — AI Robotics Track (robot learning, simulation), Hancom Innostream
Apr 2026 – Jun 2026	H-Mobility Class: Autonomous Driving Perception Track, Hyundai NGV
Sep 2022 – Dec 2022	KT AI+X Leading Talent Program, KT · Sangmyung University

SKILLS

Languages	Python (advanced), C/C++ (intermediate), Java, SQL
AI / CV	PyTorch, TensorFlow, OpenCV, ONNX, TensorRT, scikit-learn
Edge Devices	FuriosaAI Warboy, Mythic, Kneron NPU · NVIDIA Jetson Nano/Xavier · STM32 · Arduino
Tools	Ubuntu, Git, Docker, Anaconda, MATLAB, Flask, PyQt